

Decoupled Equilibrium Selection: Quantal-Response Values, CFR+ Policies, and Where Pruning Belongs in a Decomposed Solver

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Abstract

A decomposed solver for a large imperfect-information game stores two different products for every subgame: the values its parents bootstrap from, and the policy its deployment plays. This paper reports a measured finding about heads-up Red/Black, a two-player zero-sum hidden-information card game with 2.448×10^{33} perfect-recall information sets: the properties those two products need are supplied by different equilibrium selection principles, and nothing forces one method to produce both. Value correctness under composition requires uniqueness and equivariance, which an annealed quantal-response ladder supplies (colour-swap residual 3.5×10^{-9} per cell against a CFR+ baseline’s 1.67); playing strength requires sharp support, which the same ladder’s smoothed policies give up (a belief-carrying best-response probe reads a lower bound of +0.076 against the annealed profile versus +0.031 against the sharp baseline it was meant to replace). We couple the two: every subgame is re-solved with CFR+ against the quantal-response values held byte-identically fixed, and the values are deployed unchanged while only the policies are replaced. The coupled profile is the first in this program’s history that the probe cannot beat, and at 10,000 paired games per arm the probe loses to it: the reading is -0.0248 [$-0.0444, -0.0052$] against the incumbent’s $+0.0342$ [$+0.0146, +0.0538$], a difference of -0.0590 at $p = 3.0 \times 10^{-5}$, with the same direction and significance on the 6,800 fresh games that took no part in selecting the candidate. A negative lower bound is vacuous as a bound, so the claim it supports is comparative, and what it says about the probe is that this instrument has bottomed out. A contrast experiment locates where the surgery must not happen: baking a real-game-motivated pruning of a dominated action into the solved game quadruples the probe bound ($+0.1256$, $p = 1.4 \times 10^{-4}$), and a diagnostic decomposes the damage into a deployment-blindness channel (the stored profile is also the deployment’s opponent model, and a masked model assigns zero likelihood to observed reality) worth about a quarter, and intrinsic equilibrium distortion worth the rest. Pruning one’s own play is defense; pruning the world model is blindness. Every number is regenerated by a named script or artifact.

1 Introduction

Tabular solving of a large imperfect-information game begins with a decomposition. The game is cut into subgames along a public boundary, each subgame is solved against the stored values of its successors, and the deployed profile is the union of the per-subgame policies. The soundness of this bootstrap is classical [6], but it comes with a condition that is easy to state and expensive to ignore: the values a parent reads must be well-defined quantities, and in a zero-sum game only the subgame *value* is unique, while its decomposition over private types is not. Any solver that selects arbitrarily among equilibria hands each parent a slightly different game, and the selection noise composes up the bootstrap.

This paper is an experimental study of that selection problem in heads-up Red/Black, a two-player zero-sum card game of hidden information that our companion series counts, abstracts,

and solves exactly [1, 2, 3]. The study’s instrument is a belief-carrying local best-response probe that lower-bounds exploitability in the *real* game rather than in the solved abstraction, and its subject is a pair of surgeries on the solving pipeline that sit at the same joint but point in opposite directions.

The first surgery changes *how the solver selects* among the equilibria of the game it is given. Prior work in this series found that CFR+ selection freedom, composed through the bootstrap, produces value tables that violate an exact automorphism of the game by 1.67 on a $[-1, +1]$ scale, and that replacing the per-subgame solver with an annealed quantal-response (QRE) ladder repairs this to 3.5×10^{-9} [2]. The repair, however, costs playing strength: the annealed ladder’s smoothed policies measurably lose ground on the probe relative to the sharp CFR+ profile they were built to replace. The resolution this paper reports is a coupling: keep the QRE ladder’s values, frozen byte-identically, and re-solve every subgame’s *policy* with CFR+ against those fixed values. The coupled profile passes every internal acceptance check and is the first profile in the program’s history whose belief-carrying probe lower bound sits at zero.

The second surgery changes *what game the solver is given*. The solved abstraction provably rewards an action that is dominated in the real game [2], and play-time guards that prune it from the deployed policy cut the probe’s edge to a third (measured, $p = 0.014$). The natural next step, removing the action from the solved game itself so that the solve never learns it, fails badly: the masked solve’s probe bound is four times the incumbent’s ($p = 1.4 \times 10^{-4}$). A diagnostic arm separates the failure into two mechanisms with measured weights, and the asymmetry between the two surgeries is the paper’s organizing lesson. Selection can be changed freely provided the values are pinned; the solved game cannot be pruned, because the stored profile is simultaneously the deployment’s model of its opponent, and a model that assigns probability zero to actions real opponents play goes blind at exactly the moments it is being exploited.

Contributions:

- **A measured decomposition of solver error.** Value correctness under composition and policy strength at deployment are supplied by different selection principles: the QRE ladder’s uniqueness fixes the former and its smoothing damages the latter, each with its own instrument reading (Section 4).
- **A coupling architecture.** Per-subgame CFR+ re-solves against frozen QRE values, with byte-identity of the value store as an explicit invariant and the policy store as the only replaced artifact (Section 5).
- **The probe-zero result.** The coupled profile is the first the belief-carrying probe cannot beat. At $n = 10,000$ paired games the probe loses to it, $\varepsilon_G \geq -0.0248$ $[-0.0444, -0.0052]$, and the comparison against the incumbent resolves at $p = 3.0 \times 10^{-5}$ (Section 6).
- **A pruning pathology with a mechanism decomposition.** Masking a dominated action in the solved game quadruples the probe bound; a support-filter diagnostic attributes about a quarter of the regression to deployment blindness and the remainder to intrinsic equilibrium distortion (Section 7).
- **Equivariance as a meter.** Exact game automorphisms, measured and never enforced, are the acceptance instrument throughout; the one recorded enforcement blinded the only meter that could see the defect it hid (Sections 4 and 5).

Throughout, the claims are disciplined by a pre-registered distinction: a screen at 3,200

games per arm can support non-inferiority only, and any improvement claim requires a 10,000-games-per-arm extension. Both were run, in that order, and the screen’s favorable but unresolved reading ($p = 0.154$) was published as non-inferiority until the extension resolved it (Section 6).

2 The game and its abstraction

2.1 Rules sketch

Red/Black is played here heads-up with one shared 52-card deck of exactly 26 red and 26 black cards; only colour matters. Each player is dealt five cards, and hands only ever shrink. A round has four phases. In *reorder*, each player privately orders their own hand; index 0 is the top. In *bidding*, the starting player bids a pair (count, colour), asserting that at least that many cards of that colour are in play across both hands; play alternates, and each turn the active player either raises (same colour, strictly higher count) or challenges. A challenge triggers *procurement*: the bidder repeatedly flips the current top card of either player’s deck, and must reveal count cards of the bid colour before revealing a single off-colour card. The first off-colour flip loses the contest for the bidder; reaching the count loses it for the challenger. In *discard*, the loser secretly removes one card from their hand; a player whose hand reaches zero is eliminated, and the last player standing wins. Flips are public during the round they occur in and are hidden again at the next reorder; discards are anonymous, so an observer can never tell whether a previously seen card left the hand. The full rules, precise enough to implement an engine from, are in the repository’s `RULES.md` and in [1].

2.2 The real game and the abstraction

Write G for heads-up Red/Black. Exact enumeration [1] (script `real_game_size.py`) counts 5.3×10^{34} public decision histories and 2.448×10^{33} perfect-recall information sets, so no tabular method touches G directly. The same paper proves that at every round boundary the reachable public posterior over the two seats’ private types factorizes as a strategy-independent class prior times two per-seat tilt vectors, where a *class* is the public state at a procurement resolution (hand sizes, revealed-colour bounds, and the round’s public outcome data).

Definition 2.1 (The abstraction G' [1, 2]). G' is the game identical to G within each round, except that at every round boundary the joint law over type pairs is reset to the class prior; equivalently, the tilts of the factorization are replaced by 1.

Because every round removes exactly one card, the classes form a layered directed acyclic graph indexed by T' , the total number of cards in play, running from $T' = 10$ at the root down to $T' = 2$. The solved corpus has 5,721 classes. A solve proceeds by backward induction over layers: each class is solved as a two-player zero-sum subgame whose leaves are priced from the stored values of its successor classes (the *bootstrap*). For each class the store holds the game value, the per-type-pair value decomposition v_{types} (the cells parents actually read), and the behavioural policy tables.

Definition 2.2 (Round-local and real-game exploitability). $\varepsilon_{\text{round}}$ of a class solve is the exact best-response gap of its final policy *within that class’s subgame*, leaves fixed at the stored successor values. ε_G of a deployed profile is its exploitability in the real game G . All payoffs are in $[-1, +1]$, so both quantities live on a range of 2.0.

$\varepsilon_{\text{round}}$ is what a solver can drive to tolerance; ε_G is what an opponent experiences. The distance between them is the abstraction’s cost plus everything the deployment adds or repairs, and the entire methodological content of this paper is that only instruments aimed at ε_G are allowed to decide anything.

3 The instrument

3.1 The belief-carrying probe

The probe (`scripts/probe_resolver.py`, [3]) plays a local best responder (LBR [15]) against the deployed profile under *full belief carrying*: the probe maintains the exact public posterior induced by the profile’s own tables, rather than the abstraction’s reset prior, and best-responds against the profile’s actual continuation. Design elements that the numbers below depend on:

- *Shadow replay.* A shadow copy of the deployed resolver runs beside the probed one and their action distributions are asserted identical at read-back, so the probe provably measures the profile that is actually deployed rather than a reconstruction of it.
- *Paired seats and matched seeds.* Every deal is played twice with seats mirrored, under identical decision seeds; arms being compared use the same seed set (seeds pvr 0–1599, 1,600 pairs, $n = 3,200$ games per arm at screen size).
- *Wilson confidence intervals* on the probe’s win rate wr , and the conversion $\varepsilon_G \geq 2(wr - \frac{1}{2})$, so every reading is a *lower bound* on real-game exploitability. The instrument cannot certify an upper bound (Section 9).

Belief carrying is what gives the probe teeth: against the incumbent profile’s raw stored tables the bound is 0.598 per game, and full carrying by the deployed resolver cuts it to 0.078 [3], so the residual the probe measures is the part of the abstraction’s cost that deployment does not repair.

3.2 Arena blindness

The obvious cheaper instrument, head-to-head self-play between two profiles (an *arena*), is measured three times in this program to be blind to exploitability, and Table 1 collects the instances. In each case the arena read parity, or preferred the worse profile, across a difference the probe resolves decisively.

The consequence, adopted as discipline before the experiments below were run: arenas are non-regression controls only, and the probe is the verdict on every solve.

3.3 Screens and extensions

At $n = 3,200$ the probe resolves differences of roughly 0.035, and the incumbent’s entire post-guards residual is +0.031. A screen at this size can therefore support *non-inferiority* only. The program’s pre-registered rule: every candidate is screened at the benchmark’s exact seeds and size; only candidates whose point estimate moves the right way buy a 10,000-to-20,000-games-per-arm extension; and because screening several candidates and extending the best is a selection procedure, the extension, not the screen, is the honest test of an improvement claim. No sentence below claims more than its instrument resolves.

instance	arena reading	probe reading
solve-2 vs. its predecessor [2]	raw matchup <i>lost</i> by the tighter-converged solve-2	solve-2 orders of magnitude tighter on every G' meter
solve-4 vs. solve-2 ($n = 3,000$)	0.4970 [0.479, 0.515], $p = 0.76$: tie	0.1220 vs. 0.0780 (guards off, $n = 1,000/\text{arm}$)
solve-5-pruned vs. solve-2 ($n = 5,000$)	0.5078 [0.494, 0.522], $p = 0.28$: tie	+0.1256 vs. +0.0306, a $4\times$ gap resolved at $p = 1.4 \times 10^{-4}$

Table 1: Three measured instances of arena blindness. The solve-4 row reads from the artifact `probe_v10.json` and the arena record in `docs/solve-3-4-qre.md`; the last row reads from the artifacts `eval_solve-5-pruned_vs_solve2.json` and `pool_solve-5-pruned_vs_solve2.json`, both under `simulator/results/solver/`. The starkest instance is the last: a coin flip between two profiles whose belief-carrying exploitability bounds differ by a factor of four.

4 The measured decomposition

This section assembles the three measurements that motivate the coupling: CFR+ selection freedom corrupts composed values; an annealed QRE ladder repairs the values; and the same ladder’s policies give up playing strength. The section’s conclusion is one sentence, and it is load-bearing: nothing forces one method to produce both products.

4.1 CFR+ selection freedom, seen by an automorphism

Red/Black’s deck has exactly 26 red and 26 black cards and no other card attribute exists, so swapping the colours is an automorphism of G and of G' (the class prior is colour-symmetric). The value of a class must equal the value of its colour-swapped image with types remapped, exactly, for any equilibrium. The solver never enforces this, which is what makes it a meter.

Measured on the CFR+ solve of record, solve-2: of 2,971 colour orbits, 2,206 violate, with worst v_{types} cell error 1.67 on the $[-1, +1]$ scale, while the same classes report $\varepsilon_{\text{round}}$ between 5×10^{-5} and 10^{-4} (script `check_symmetry.py`, artifact `symmetry_full.json`). The defect is invisible to every convergence meter by construction, because each class solve is internally consistent with the corrupted leaves it was handed. A discriminating experiment (script `symmetry_discriminate.py`, artifact `symmetry_discriminate.json`) exonerates the game logic entirely: re-solving violating orbits with leaves forced symmetric lands every twin equivariant to machine precision, while raw store leaves differing by at most 10^{-4} produce cell jumps of 0.14–0.45. The mechanism is equilibrium-selection freedom. The subgame value is unique but v_{types} is not; CFR+ selects arbitrarily among equilibrium decompositions, self-swap classes seed $\sim 4 \times 10^{-3}$ of cell asymmetry on perfectly symmetric inputs, an ill-conditioned cell map amplifies input differences roughly 1000 $\times$, and parents inherit genuinely different games [2].

Remark 4.1 (An enforced invariant is a blinded meter, 2026-08). An earlier solve in this series enforced the seat-mirror symmetry by construction, after which the mirror gap read zero because it was forced to. The colour swap, which nothing enforced, later read violations of the same ~ 0.11 magnitude on the enforced solve: the defect had been hidden, not fixed [2]. Every symmetry in this paper is measured and never imposed, and the coupling’s acceptance checks (Section 5) inherit that rule unchanged.

4.2 The annealed QRE ladder

The repair is to make the per-class selection unique. The QRE ladder (driver `qre_solve.py`, kernel mode `QRE`) replaces CFR+ within each class by magnetic mirror descent [7] on an entropy-regularized game: per information-set row,

$$\ell \leftarrow \frac{\ell + \eta q/c}{1 + \eta\tau},$$

with q the counterfactual action value normalized by counterfactual reach c , step size $\eta = 4$, and the temperature τ annealed down a fixed geometric ladder from 0.3 to $\tau_f = 3 \times 10^{-3}$, taking the last iterate. For every $\tau > 0$ the regularized game has a unique equilibrium, continuous in its leaf inputs, so the composition problem of Section 4.1 is solved by construction rather than by tolerance. Measured on the full re-solve (solve-4): colour equivariance holds on all 2,971 orbits with worst residual 3.5×10^{-9} per cell, against the CFR+ baseline’s 1.67 (`check_symmetry.py`; [2]).

Cost matters because the ladder must be re-runnable. A value-based stopping rule (stop a rung when the class value stops moving, rather than on an iteration budget) cuts the ladder roughly 30 $\times$, from 40,406 iterations per class to about 1,300, and was validated on the full 360-orbit policy-equivariance sweep rather than a pilot: the divergent-twin tail shrinks from 22 to 6 orbits with overall ε quality unchanged (worst-twin median 0.0030 before and after; artifact `deps_value_stoprule_full.json`). The fast ladder is what makes every experiment in Sections 5 and 7 affordable.

4.3 What the ladder costs: smoothed policies are weak

The QRE profile’s values are right and its play is not. In the arena the annealed solve ties the CFR+ incumbent (0.4970 [0.479, 0.515], $p = 0.76$, $n = 3,000$; `docs/solve-3-4-qre.md`), which by Section 3.2 decides nothing. The probe decides: with play-time guards on both profiles, the belief-carrying bound against the QRE profile is +0.076 [+0.014, +0.137] at $n = 1,000$ (`s5_probe_solve4_guards.json`) against the CFR+ incumbent’s +0.031 [−0.004, +0.065] at $n = 3,200$ (`s5_probe_solve2_guards.json` and `_ext.json`). The ladder’s full-support, temperature-smoothed policies concede ground that the sharp CFR+ policies do not, at the same time as its values pass an exactness test the CFR+ values fail by nine orders of magnitude.

The decomposition is now measured on both sides. Value correctness under composition needs uniqueness and equivariance; deployment strength needs sharp support. One selection principle supplies each, and nothing forces one method to produce both.

5 The coupling

5.1 Architecture

The coupled solve (store `solve-6-cfrpolicy`, driver `scripts/policy_pass.py`) re-solves every one of the 5,721 classes with CFR+ at tolerance 10^{-5} (achieved median $\approx 10^{-5}$) against leaf prices taken from the QRE ladder’s value store and held *fixed*: the CFR+ pass reads solve-4’s v_{types} cells and never writes values back. The deployed value artifact is solve-4’s `values.pkl.gz`, byte-identical (sha256-verified), plus new policy tables. Two design decisions carry the result.

Frozen values are load-bearing. CFR+ within a class converges to its own arbitrary equilibrium decomposition, and letting its values flow back into the store would re-import the exact

selection noise of Section 4.1. The size of the freedom being discarded was measured. Over the full 5,721-class pass, CFR+’s own v_{types} cells drift from the held QRE prices by as much as 1.4244; on a stratified 90-class re-solve sample the per-class maximum cell drift has median 0.0293, 95th percentile 0.5822, and maximum 0.7324, while the entry-weighted class values of the same solves agree with the held prices to a median of 1.0×10^{-4} and never differ by more than 1.1×10^{-3} (`coupling_value_drift.py`, artifact `coupling_value_drift.json`). Four orders of magnitude separate the cell drift from the value drift: both solvers agree about what the class is worth and disagree about how to decompose that worth across type pairs, which is v_{types} non-uniqueness doing exactly what Section 4.1 says it does.

One price, every consumer. Internal consistency is a stated invariant: the value store every parent bootstraps from, the prices the deployed resolver’s leaf evaluations use, and the cells the analysis tools display are one artifact, and the sha256 identity of `values.pkl.gz` with its solve-4 source is checked as an acceptance condition, so no consumer can drift onto a second price.

5.2 Acceptance checks

The feared cost of the coupling was the return of per-class selection freedom in the *policies*. The policy-equivariance sweep (script `check_policy_symmetry.py`, artifact `policy_symmetry_solve-6-cfrpolicy.json`) measures the policy distance d_ϵ across all 360 audited colour orbits: 360/360 records, median 1.2×10^{-9} , maximum 0.000. The cost never materialized, and the reason is worth recording: deterministic CFR+ is an equivariant map, so over colour-equivariant inputs (the frozen QRE prices) it returns colour-equivariant policies. The selection noise of Section 4.1 was never CFR+ acting alone; it was CFR+ composing its own corrupted inputs up the bootstrap, and pinning the inputs removes the seed.

The remaining checks are shaped by the incumbent. Tripwire telemetry (script `tripwires.py`, artifact `tripwires_solve-6-cfrpolicy.json`) reads incumbent-shaped: the coupled profile is unmasked and plays the G' -equilibrium rate of the real-game-dominated action natively, and play-time guards keep handling it at deployment, a division of labor that Section 7 justifies by measurement. Compute for the full pass: 67,982 seconds of wall clock, about 19 hours including a five-hour suspension, on one 10-core Apple M-series machine with 32 GB of memory (`policy_pass.py` run record in `simulator/logs/solve-5-journal.md`).

6 Verdict

Table 2 is the verdict of record, at the benchmark’s exact seeds, paired deals, and mirrored seatings. Every contributing run logged zero resolver breaks and zero fallbacks.

The coupled profile is the first in this program’s history whose belief-carrying probe cannot beat it. At screen size the probe’s win rate was indistinguishable from a coin flip and the difference against the incumbent was -0.0356 , $z = -1.43$, $p = 0.154$, favorable in direction and unresolved, exactly as the pre-registration of Section 3.3 predicts for a true difference of this magnitude. The extension resolves it. Pooled at 10,000 games per arm the difference is -0.0590 , $z = -4.17$, $p = 3.0 \times 10^{-5}$, and the claim of record moves from non-inferiority to improvement.

The extension alone carries that claim, which matters because the screen selected this candidate and screening several candidates then extending the best is a selection procedure. On the 6,800 fresh games that took no part in the selection, the difference is -0.0700 , $z = -4.08$, $p = 4.5 \times 10^{-5}$, slightly larger than the pooled figure. Nothing here depends on reusing the screen.

arm	stage	probe wr	ε_G lower bound
solve-6-cfrpolicy (coupled)	screen	0.4975	-0.0050 [-0.0396, +0.0296]
solve-2, guards on (incumbent)	screen	0.5153	+0.0306 [-0.0040, +0.0652]
solve-6-cfrpolicy (coupled)	extension	0.4829	-0.0341 [-0.0578, -0.0104]
solve-2, guards on (incumbent)	extension	0.5179	+0.0359 [+0.0121, +0.0596]
solve-6-cfrpolicy (coupled)	pooled	0.4876	-0.0248 [-0.0444, -0.0052]
solve-2, guards on (incumbent)	pooled	0.5171	+0.0342 [+0.0146, +0.0538]

Table 2: Screen ($n = 3,200$ per arm, seeds pvr 0–1599), pre-registered extension ($n = 6,800$, fresh seeds pvr 1600–4999), and the two pooled ($n = 10,000$, seeds 0–4999). All paired. Artifacts `probe_solve-6-cfrpolicy_a.json/_b.json` and `_ext_a.json/_ext_b.json`; benchmark `s5_probe_solve2_guards.json`, `_ext.json` and `_ext2.json`. Pooling is by `pool_probe.py`, whose two artifacts are named in Section 10.

The sign carries the more interesting reading. The probe does not merely fail to beat the coupled profile, it loses to it, 0.4876 against a fair 0.5 ($p = 0.014$ pooled, $p = 0.005$ on the fresh games). A negative lower bound is vacuous, since $\varepsilon_G \geq 0$ holds by definition, so the honest statement of the absolute result is that the belief-carrying probe yields no positive lower bound on this profile’s exploitability at $n = 10,000$. That is a fact about the instrument rather than a proof of near-optimality, and it is the sense in which this program now lacks a probe strong enough to price its own best profile. The instrument is otherwise healthy: the same probe at the same seeds with the same guards reads +0.0342 with a confidence interval clear of zero against the incumbent, which is what makes the comparative claim well posed even though the absolute one is not.

Because the two arms were measured on different machines, both stores were fingerprinted file by file on each before pooling. The coupled store is bit-identical across machines, and the incumbent’s differs only in a Finder metadata file and in ablation directories no probe reads. `pool_probe.py` compares the runs’ configuration keys but not the stores’ contents, and the `store` field it carries is a filesystem path, so this check is external to the tooling by construction.

The arena control is consistent with non-regression and decides nothing further: against the incumbent, 0.5096 [0.496, 0.523], $p = 0.18$ guards-off and 0.5048 [0.491, 0.519], $p = 0.51$ guards-on, at $n = 5,000$ per mode (artifact `eval_solve-6-cfrpolicy_vs_solve2.json`). Per Section 3.2 the arena’s role here is only to rule out gross breakage, and it does.

7 The contrast arm: pruning the solved game

The coupling changed how the solver selects. The contrast arm changed what game it solves, and its failure is what locates the boundary between the two kinds of surgery.

7.1 Background: a dominated action the abstraction rewards

The solved abstraction rewards an action that the real game punishes: at some procurement information sets, flipping one’s own known off-colour card (a *suicide flip*, guaranteed to lose the contest) is simultaneously optimal in G' and dominated in G , because the belief reset launders the self-revelation the flip would cost in the real game [2]. The G' equilibrium plays it with probability up to 0.97 on-path. Two independent solvers corroborate the artifact: the CFR+

era measured $p = 0.97$ with 0.0499 of the affected class’s value attached, and the QRE ladder, a different engine and update rule, returns $p = 0.96$ and 0.0497 (`tests/test_solver_dominance.py`; `simulator/logs/solve-5-journal.md`).

Play-time guards, which prune the action from the deployed policy’s own play and touch nothing else, are measured to help: on the incumbent the probe bound falls from +0.0919 [+0.0573, +0.1262] guards-off to +0.0306 [−0.0040, +0.0652] guards-on, $z = 2.45$, $p = 0.014$ at $n = 3,200$ per arm (artifacts `s5_probe_solve2_base.json`, `s5_probe_solve2_guards.json`, with `_ext` extensions). Guards cut the probe’s edge to a third. The tempting inference is that removing the action from the solved game itself, so no store ever learns it, should do at least as well.

7.2 The masked solve

The masked solve (store `solve-5-pruned`) runs the identical fast QRE ladder with the suicide flip made illegal inside every class subgame. Provenance is read back from the finished policies rather than from the configuration that requested them: 2,209/2,209 canonical class-solves masked, 14,777,285 policy rows locked, worst residual mass on any locked row exactly 0 (append-only record `solve-meta.jsonl`; colour-swap and d_ϵ checks in `symmetry_solve-5-pruned.json` and `policy_symmetry_solve-5-pruned.json`). The deployment masks its resolves the same way, so the deployed strategy space is the solved one. The verdict, at the benchmark’s seeds and size:

arm	probe wr	ε_G lower bound
solve-5-pruned (masked, guards on)	0.5628	+0.1256 [+0.0911, +0.1598]
solve-2, guards on (benchmark)	0.5153	+0.0306 [−0.0040, +0.0652]

Difference +0.095, $z = 3.81$, $p = 1.4 \times 10^{-4}$; the intervals do not overlap (artifact `pool_solve-5-pruned_vs_solve2.json`). Baking into the solve the same removal that helped at play time *quadrupled* the probe’s bound.

7.3 The diagnostic, and the two mechanisms

The first suspect surfaced in telemetry: 806 of 3,200 probe games against the masked profile broke the deployed resolver’s opponent filter, against 0 of 3,200 in the benchmark. A *break* is the filter observing an action to which its model assigns zero likelihood, after which that game softly degrades to static tables. The masked store makes this structural. The stored profile is also the deployment’s opponent model; the masked model says the suicide flip has probability zero; the probe, an external best responder correctly not subject to the mask, plays it when the reset artifact makes it attractive; and the filter then cannot explain observed reality.

The diagnostic arm re-probes the *same* masked store with the filter’s opponent floor extended to structural zeros (`--opp-support-filter`), same seeds, same size. Breaks fall from 806/3,200 to 0/3,200, and the bound falls from +0.1256 to +0.0931 [+0.0585, +0.1275], which is still decisively worse than the benchmark ($z = 2.50$, $p = 0.012$; artifact `pool_solve-5-pruned-osf_vs_solve2.json`). The decomposition:

- *Deployment blindness, about a quarter of the regression.* Eliminating the break channel recovers $0.126 \rightarrow 0.093$. The mechanism is general: whenever the stored profile doubles as the deployment’s opponent model, pruning the store creates a model that zero-likelihoods actions real opponents play, and inference dies exactly when it is needed.

- *Intrinsic equilibrium distortion, the remaining three quarters.* The masked G' equilibrium prices every surrounding decision against an action set no real opponent is restricted to. The removed action is genuinely G' -optimal at $p = 0.96$ – 0.97 , and masking it surrenders about 0.05 of the affected class’s value; the whole strategy shifts around that surrender, and the shifted strategy is what the real game exploits.

One caveat is carried rather than hidden: the support filter also coarsens opponent inference generally, so the diagnostic is not a surgical break-only ablation; but with breaks at exactly zero and the bound within noise of the verdict’s, no reading of it rescues the mask. The closing statement of the contrast arm, earned by three measurements (guards help at $p = 0.014$; the mask hurts at $p = 1.4 \times 10^{-4}$; the diagnostic splits the damage): pruning your own play is defense, and pruning your model of the world is blindness. A dominated action leaves the deployed policy at play time, or it does not leave at all.

8 Related work

Decomposition and re-solving. Subgame decomposition with stored summaries is due to Burch et al. [6] (CFR-D), whose soundness argument carries an equilibrium-selection invariance condition: the stored quantities must mean the same thing to every subgame that reads them. The QRE ladder of Section 4.2 can be read as a constructive way to satisfy the uniqueness that decomposition wants, and the coupling keeps that construction while discarding the ladder’s policies.

Solver-dependent selection. The nearest neighbor to Section 4.1 is the recent characterization of solver-dependent selection on zero-sum Nash polytopes [8], which studies how different algorithms select different points of the equilibrium set. Our contribution is complementary and empirical: a measured account of what that freedom costs when selections *compose* through a value bootstrap (1.67 per cell), and a deployed architecture that pins the selection where composition needs it and frees it where deployment does.

MMD and QRE. Sokota et al. [7] introduce the magnetic mirror descent update we use and anneal it toward a Nash *policy*. Our reading of the same ladder inside a decomposition is different: the ladder’s deliverable is the *values*, whose uniqueness and equivariance are what composition consumes, and we measure that its annealed policy should be discarded (Section 4.3).

Learned value functions. DeepStack [9], value functions for depth-limited solving [10], ReBeL [11], and Student of Games [12] price subgame leaves with *learned approximate* value functions for the sake of scale. The present system is the small-scale exact counterpart: both selection principles are exact solvers, chosen from a measured error decomposition, with the values frozen byte-identically rather than approximated. What transfers in either direction is the coupling question itself, since any decomposed agent must decide which product each component is trusted for.

Abstraction and pruning pathologies. Waugh et al. [13] showed that refining an abstraction can hurt the resulting equilibrium. Section 7 is a *pruning* pathology with a mechanism decomposition: restricting the solved game’s action set with a real-game-motivated, provably dominated removal quadrupled measured real-game exploitability, and a quarter of the damage flows through a channel we have not found documented, the deployment’s use of its own store as its opponent model. Regret-based pruning [14] sharpens the distinction from the other side: it prunes *computation* while provably preserving the solution, whereas the mask prunes the *game*, and the two must not be conflated under the one word.

We characterize the system’s novelty as a *measured coupling*: the components are standard, and the contribution is the error decomposition that says which component supplies which product, with an instrument reading behind each arrow.

9 Limitations

One game. Every measurement is from heads-up Red/Black. The mechanisms are stated generally (selection noise composing through a bootstrap; a store doubling as an opponent model), but their weights (1.67; a quarter versus three quarters) are this game’s.

A lower-bound instrument. The probe certifies ε_G from below only. The companion certificate program pursued an upper bound and reports its obstruction honestly [3]: leaf-price dishonorability of 0.11–0.26 per re-solve is intrinsic to depth-limited resolving from any stored belief-reset surface, and no upper bound below the trivial 2.0 is claimed. Probe-zero therefore means the instrument cannot beat the profile, and does not mean the profile is unbeatable. The coupled profile has now passed the point where this instrument says anything absolute about it at all: the bound it returns is negative, hence vacuous, and only the comparison against a profile the probe can still beat remains informative. Pricing the coupled profile absolutely requires a stronger probe, and building one is open.

Temperature bias in the values. The deployed values are the QRE ladder’s at $\tau_f = 3 \times 10^{-3}$, and carry its bias: the root value of G' differs across the CFR+ and QRE solvers at the third decimal (+0.0245 versus +0.0190; [2], docs/solve-3-4-qre.md). Composition consumes the values’ uniqueness, not their unbiasedness, but any consumer quoting a cell as ground truth inherits the τ_f floor.

Provenance. This is a single-author study of a bespoke domain. The mitigations are structural: every number in this paper names the script or artifact that regenerates it, the game rules are specified to engine-implementable precision [1], and the probe’s shadow-replay and read-back assertions are designed so that the measured object is provably the deployed one.

10 Reproducibility

The series’ convention applies unchanged: no number without a script. Game counts regenerate from `real_game_size.py` and `state_counts.py` [1]; solver measurements regenerate from the scripts under `simulator/scripts/` in the game repository (`qre_solve.py`, `policy_pass.py`, `check_symmetry.py`, `check_policy_symmetry.py`, `tripwires.py`, `probe_resolver.py`, `pool_probe.py`), whose JSON artifacts live under `simulator/results/solver/` with the store name in the filename (`probe_solve-6-cfrpolicy_*.json`, the verdict’s two pooled artifacts `pool_solve-6-cfrpolicy_vs_solve2_full.json` and `pool_solve-6-cfrpolicy_vs_solve2_extonly.json`, `policy_symmetry_*.json`, `tripwires_*.json`, and `eval_*.json`). Probe seed sets are named in every artifact (pvr ranges), deals and decision seeds are deterministic given the seed, and paired arms share seed sets, so any reported comparison re-runs to the same games. The value store’s identity claims are sha256 hashes, checkable offline. The papers, their scripts, and the counting validators are public at github.com/SwiftlyNoah/redblack-research.

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